

Mohamed Abderrahmane Allouche

Full-Stack Software Engineer

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PROFESSIONAL SUMMARY

Full-Stack Software Engineer with 3 years of experience building scalable web applications, backend AI services, and cybersecurity SaaS platforms using TypeScript, React, Python (FastAPI/Django), and PostgreSQL. Skilled in architecting 20+ secure RESTful API endpoints, full-duplex WebSocket protocols, multi-tier authentication systems (JWT/RBAC), and asynchronous background task pipelines with Celery. Experienced across the full software lifecycle from relational schema design and AI model integration to Docker containerization and cloud deployment. Ranked 3rd in Computer Science cohort of 120+ graduates.

TECHNICAL SKILLS

- **Languages:** TypeScript, JavaScript (ES6+), Python, SQL, Java, PHP, HTML5/CSS3, C#
- **Frontend:** React.js, Angular, Next.js, Redux Toolkit, Tailwind CSS, Bootstrap, Vite, Responsive UI
- **Backend:** FastAPI, Node.js, Express.js, Django, Pydantic, SQLAlchemy, Celery, .NET
- **Databases:** PostgreSQL, Supabase, MongoDB, MySQL
- **APIs & Auth:** REST APIs, WebSockets, JWT, OAuth 2.0, Role-Based Access Control (RBAC), Argon2
- **DevOps & Tools:** Docker, AZURE, Git, GitHub, Vercel, Railway, Render, Linux, DLT
- **Security:** OWASP Top 10, Wazuh, Burp Suite, Metasploit, Nmap, Wireshark, Kali Linux

WORK EXPERIENCE

FlyRank Corp. (FlyRank AI)

Back-End AI Engineering Intern

Remote, USA

July 2026 – August 2026

- Engineered backend RESTful API endpoints in Python and FastAPI during an intensive engineering program to support AI-driven features and model integration workflows.
- Architected asynchronous data-processing pipelines leveraging background task queues, Pydantic request validation, and structured JSON serialization for model outputs.
- Partnered with cross-functional engineering teams to formulate clean backend architecture, relational database schemas, API contracts, and technical documentation.

SocShield DZ

Full-Stack Developer

Hybrid, Algeria

August 2023 – Present

- Constructed modular security-monitoring dashboards across 15+ data views using React, TypeScript, and Redux Toolkit, managing complex asynchronous application state.
- Designed and deployed 20+ RESTful API endpoints using FastAPI and PostgreSQL, implementing Pydantic validation schemas and SQLAlchemy for relational database querying.
- Orchestrated token-based JWT authentication and Role-Based Access Control (RBAC) across 4 user privilege tiers, enforcing strict authorization policies across client portals.
- Enforced OWASP Top 10 security standards across application layers, mitigating injection and XSS risks through parameterized database queries and structured code reviews.

PROJECTS

LeakShield | *FastAPI, SQLAlchemy, PostgreSQL, Celery, React*

- Spearheaded backend infrastructure for an automated OSINT data-breach monitoring platform, orchestrating Celery workers and PostgreSQL to index exposed digital assets.
- Created responsive React dashboards with real-time search, multi-criteria filtering, and asset-exposure synchronization backed by Supabase.

Devfolio – AI Portfolio Builder | *FastAPI, React, PostgreSQL, Pydantic, SQLAlchemy*

- Authored a full-stack portfolio platform using FastAPI, React, and PostgreSQL, enabling users to upload resume files and generate customized, responsive personal websites.
- Integrated third-party GitHub OAuth 2.0 and local authentication secured with memory-hard Argon2 password hashing and JWT access tokens.

Pulse-Chat | *React, TypeScript, FastAPI, WebSockets*

- Engineered a full-duplex real-time messaging application using React, TypeScript, FastAPI, and WebSockets, sustaining 100+ concurrent connections in testing.
- Designed multi-room messaging architecture featuring authenticated WebSocket handshakes, connection lifecycle monitoring, and automated client reconnection logic.

EDUCATION

WSB University

B.Sc. in Cybersecurity

Kraków, Poland

Expected: October 2026 – July 2030

Université Dr. Moulay Tahar

B.Sc. in Computer Science (Ranked 3rd in Cohort of 120+)

Saïda, Algeria

September 2023 – June 2026

Relevant Coursework: Data Structures & Algorithms, Web Development, Web Security, Operating Systems, Linux Systems, AI & Machine Learning, Language Theory & Compilation.

ACHIEVEMENTS & CERTIFICATIONS

- **Achievements:** 2nd Place Winner – Dev Rush 1.0 National Programming Competition (2025); National Finalist FORSA Algérie Télécom Hackathon & Datathon (2025)
- **Certifications:** Google Cybersecurity Professional Certificate (2024); NVIDIA Fundamentals of Deep Learning (2025)[cite: 2]; Huawei HCIA-Datacom (2025); Simplilearn CCNA Network Associate (2025)
- **Languages:** Arabic (Native), English (Professional IELTS Academic 6.5, EFSET C1), French (Advanced TCF C1)